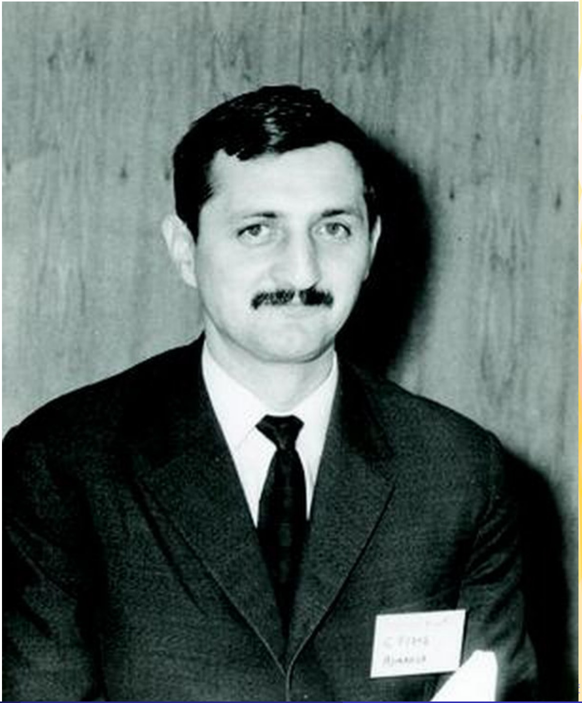
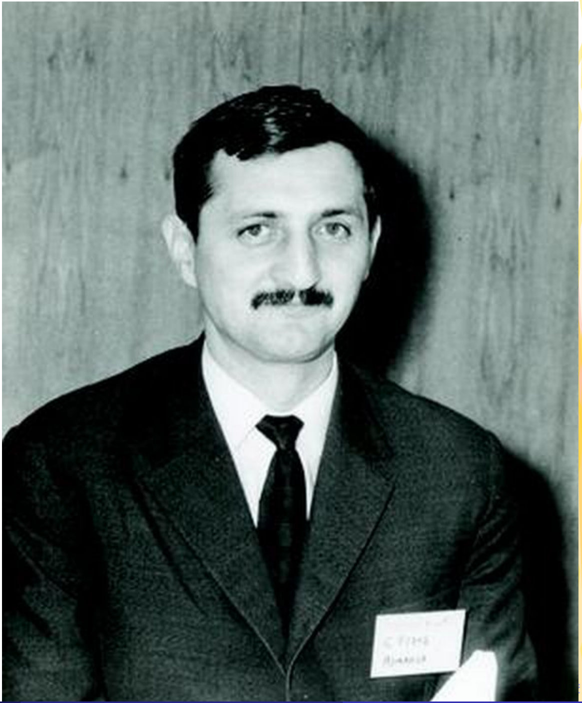
Foias, Agler and the future

Pascoe, Drexel

June 29, 2026

or, the Gelfand theory.





Theorem

Let $\mathcal{H}$ be a Hilbert space. Let $T \in \mathcal{H}$ such that $\|T\| < 1$. (We are assuming strict for ease of discussion.) There is a Hilbert space $\mathcal{K}$, as isometry $P : \mathcal{H} \to \mathcal{K}$, and $U \in \mathcal{U}(\mathcal{K})$ such that for every bounded analytic function f on $\mathbb{D}$.

$$f(T) = P^* f(U) P.$$

That is, every contraction dilates to a unitary.

Sz-Nagy Dilation

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Schaeffer Form

$$\begin{bmatrix} T & 0 & 0 & 0 & \cdots \\ \sqrt{1-T^*T} & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

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Automorphic Nelson's trick

Let $b_r(z) = \frac{z-r}{1-rz}$. Let T be a matrix. Write $T = U^*\Sigma V$.
$\hat{T}(z) = U^* b_T(z) V$ is obviously an isometric or unitary dilation depending on whether you interpret over Hardy space or L^2. Minimality (Bickel-P.-Tully-Doyle.)

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Characteristic function

Write $D_X = \sqrt{1 - X^*X}$. $\|T\| < 1$. The characteristic function is

$$\theta_T(z) = -T + zD_{T^*}(1 - zT^*)^{-1}D_T.$$

(Obviously inner. Sz.-Nagy-Foias theory. $-\theta_T$ dilation.)
Problem: Link the characteristic function to the Nelson Dilation.

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From Ciprian's obituary

"In 1978, Ciprian defected while he was at a mathematics conference in Helsinki, Finland. The stories our parents told us about that moment in his life made it sound very exciting and very scary at the same time. With the help of his French colleagues, he left his hotel room in the morning and took several taxis so that the Romanian secret service could not follow him. Then, his French colleagues put him on the plane and when my father got off the plane in France, without a visa, he asked for political asylum. The next year, while he was in France alone, and while we and my mother were in Romania, were torture as there was no certainty that the Communist regime would let us to leave Romania to join him abroad. But, many people helped, including the French government, the Indiana senators, as well as the mathematician daughter of the Romanian president (Nicolae Ceausescu), who had been one of Ciprian's students in Romania. After one year, we gained permission to leave Romania. We left our extended family behind, but were given the chance for a better life in United States."

Happy 250th from the United States. 

Part II: Agler

Have you seen this man? Missing since 2020...

My life may be encapsulated by one of Graham Greene's "entertainments" titles: 'Loser Takes All'. Since I was thrown out of highschool for political reasons, I was free to study on my own and develop my own ways of thinking. - Gelfand

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Agler has a very particular "philosophy of math." The Hahn-Banach theorem has many forms, and, thus, at the core, there is no Hahn-Banach theorem.

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Pick interpolation

Theorem (Pick)

Let $z_1, \ldots, z_n \in \mathbb{D}$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$. The matrix $[\frac{1-\lambda_i\overline{\lambda_j}}{1-z_i\overline{z_j}}]$ is positive semidefinite if and only if there is an $f : \mathbb{D} \to \overline{\mathbb{D}}$ analytic such that $f(z_i) = \lambda_i$.

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Suppose the matrix $A = [\frac{1-\lambda_i\overline{\lambda_j}}{1-z_i\overline{z_j}}]$ is positive semidefinite.

Thus, $\frac{1-\lambda_i\overline{\lambda_j}}{1-z_i\overline{z_j}} = \langle u_i, u_j \rangle$ for some vectors.

Now,

$$1 + \langle z_i u_i, z_j u_j \rangle = \lambda_i \overline{\lambda_j} + \langle u_i, u_j \rangle.$$

So

$$\begin{bmatrix} \lambda_i \\ u_i \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} 1 \\ z_i u_i \end{bmatrix}$$

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Theorem (Engineering)

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Dilation on the Annulus

Let T be an operator. We say K is a spectral set for T if for every analytic function f on a neighborhood of K we have $\|f(T)\| \leq \|f\|_{L^\infty(K)}$.

Theorem (Agler, Dritschel-McCullough)

An annulus is a spectral set for T if and only if there is a normal dilation of T with spectrum in the boundary of the annulus.

Fails for triply connected domains.
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Part III: The story continues

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Then for every analytic function f on the corresponding annulus,
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Holds on various other multivariate domains (Tomar.)

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In full generality, a free noncommutative function is a functor from C^E to C^F which respects natural transformations. (Bushelli)
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Positive definite functions

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Theorem (Damase-P.)

A function $f : X \to \mathbb{R}$ satisfies "if $[a_{ij}]_{ij}$ is a partially defined psd matrix with entries in X, then $[f(a_{ij})]_{ij}$ is a partially defined psd matrix" if and only if $f(t) = \sum a_n t^n, a_n \geq 0$.

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Dance with the ugly duckling not the boar's slop.